SPARQL patterns for the file-type assay

Base templates | 8 October 2026

These are the query patterns behind typeAssay.pdf, typeAssay-v2.pdf, and the later pass that used encodingFormat on URLs the filename rule left unknown. Each catalog was queried separately. The file-type decision was made after the query, in Python. It is not part of the SPARQL.

Endpoints: https://qlever.geocodes-aws.earthcube.org/graphspace/deepoceans , .../ecoforecast , and .../earthsurface . facetsearch: https://qlever.geocodes-aws-dev.earthcube.org/graphspace/facetsearch .

1. Count distributions and content URLs

Used for the object-count and distinct-URL rows. No LIMIT.

```
PREFIX schema: <https://schema.org/>
SELECT (COUNT(DISTINCT ?dist) AS ?downloads)
       (COUNT(DISTINCT ?url) AS ?urls)
WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload .
    OPTIONAL { ?dist schema:contentUrl ?url . }
  }
}
```

2. List distinct content URLs

This is the URL set that was classified from the filename. Version 1 sent LIMIT 200000. Version 2 dropped the LIMIT. Both returned the same distinct URLs, and the counts matched pattern 1.

```
PREFIX schema: <https://schema.org/>
SELECT DISTINCT ?url
WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload ;
          schema:contentUrl ?url .
  }
}
```

A short LIMIT was used only to inspect example URLs before the full pull. That sample is not a separate template.

3. List predicates on a distribution

Used once per catalog to see which URL and format properties were actually present (contentUrl, url, encodingFormat, name, description). LIMIT 30 was only a display cap on the predicate list.

```
PREFIX schema: <https://schema.org/>
SELECT ?p (COUNT(*) AS ?n)
WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload ;
          ?p ?o .
  }
}
GROUP BY ?p
ORDER BY DESC(?n)
```

4. Aggregate encodingFormat

Used to see the format values before joining them back to unknown URLs. No LIMIT.

```
PREFIX schema: <https://schema.org/>
SELECT ?fmt (COUNT(DISTINCT ?url) AS ?urls)
            (COUNT(DISTINCT ?dist) AS ?downloads)
WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload ;
          schema:contentUrl ?url ;
          schema:encodingFormat ?fmt .
  }
}
GROUP BY ?fmt
ORDER BY DESC(?urls)
```

One ecoforecast check was this same pattern with FILTER(LCASE(STR(?fmt)) = "application/x-parquet"). That confirmed 337 distinct contentUrls, which matched the later pair join.

5. Pair each content URL with its encodingFormat

This is the query that resolved unknowns. A URL was given a type only when every file-type value on that URL mapped to the same type. Access methods, octet-stream, and conflicting types were not mapped.

```
PREFIX schema: <https://schema.org/>
SELECT DISTINCT ?url ?fmt
WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload ;
          schema:contentUrl ?url ;
          schema:encodingFormat ?fmt .
  }
}
```

6. Read name and description

Used only on URLs that pattern 5 did not resolve. The base shape is a distribution plus its name and description. Probes added a URL filter, or a REGEX on the concatenated text, and sometimes LIMIT. No name or description stated a single file type that encodingFormat had missed, so none of these probes changed a count.

```
PREFIX schema: <https://schema.org/>
SELECT DISTINCT ?url ?name ?desc
WHERE {
  GRAPH ?g {
    ?dist a schema:DataDownload ;
          schema:contentUrl ?url .
    OPTIONAL { ?dist schema:name ?name }
    OPTIONAL { ?dist schema:description ?desc }
    FILTER(CONTAINS(STR(?url), "HOSTNAME"))
  }
}
```

HOSTNAME stood in for a residual host, such as ssdb.iodp.org or doi.org. A second form replaced the URL filter with BIND(CONCAT(COALESCE(STR(?name), ""), " ", COALESCE(STR(?desc), "")) AS ?text) and FILTER(REGEX(?text, "text/csv|application/zip|GeoTIFF|NetCDF", "i")).

What was not queried

Suffix classification (.csv, .tif, files=name.gpkg, unknown) was client-side. The four catalogs were not joined in one query. schema:url was counted in the predicate inventory but was not used as the file URL. schema:encodingFormat was not used in version 1 or version 2; it entered only in the unknown-resolution pass.